

Assignment 1

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Cryptocurrency is a type of digital money that allows people to send and receive funds over the internet without needing banks. Prices of cryptocurrencies like Bitcoin and Ethereum can change very quickly, within minutes.

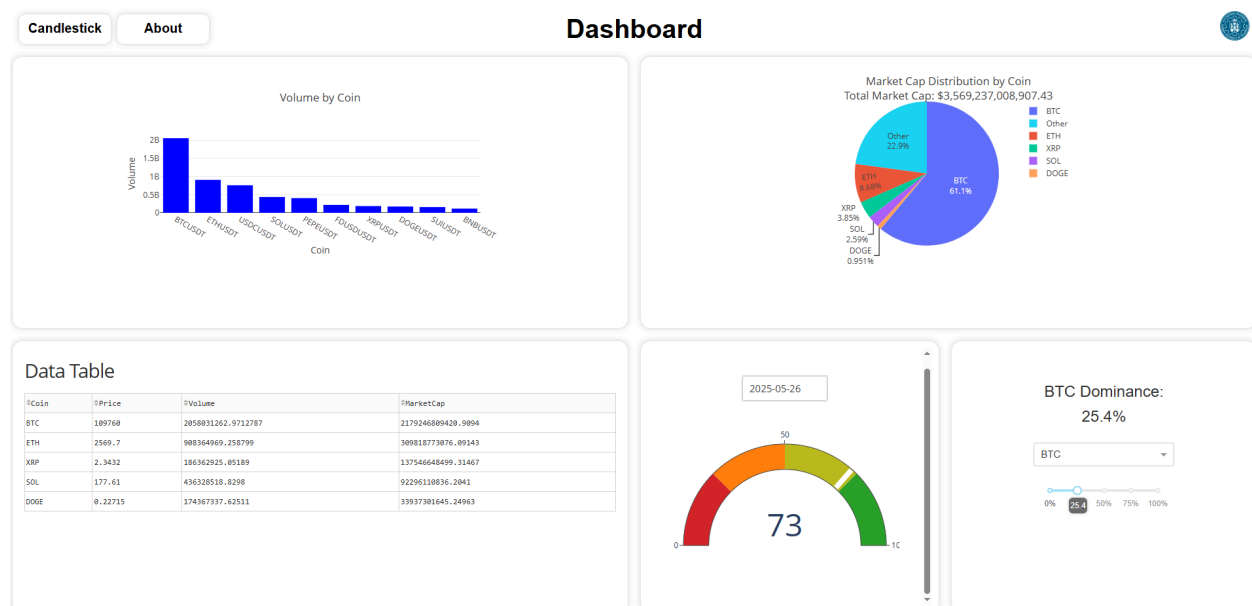
Because of this volatility, it's important to track your crypto investments. A crypto dashboard is a tool that collects all crypto data in one place. It shows prices, portfolio value, gains or losses, and trends in real-time. This makes it easier to make informed decisions, stay updated, and manage your investments efficiently.



Example of bad design: my friends' Crypto Dashboard

This dashboard was designed to visualize cryptocurrency data using multiple charts (bar, pie, gauge) and a data table.

Its main goal is to summarize market information - volume, price, market cap, and BTC dominance - in a single dashboard.



Why is it bad?

- Raw, Unformatted Data
 - Market cap and volume values include excessive decimals, making them hard to read (e.g., 2179246809420.9094)
 - No currency symbols or thousand separators
- Bad Visual Aspects
 - All charts are given equal space, even though some (like data table) are more relevant
 - The layout is fragmented - each chart is in a large, empty card, leading to inefficient use of space
 - The large "Dashboard" title and oversized cards waste valuable screen space
 - The user's attention isn't guided - they must manually scan everything, it's not clear where to look at
 - Important metrics (like BTC dominance) don't visually stand out - everything has the same weight.
 - Lack of contrast and visual harmony
 - Font sizing and spacing are inconsistent.
 - No consistent color theme across charts
- Low interactivity

- The dashboard is static - no tooltips, filtering, sorting, or clickable elements
- Data does not refresh in real time
- There is no chart inside data table that allows to compare trends
- Navbar is broken, text switches places after clicking one of the options

The dashboard is a bad interface because it presents raw, unformatted data with excessive decimals and no currency symbols, making numbers hard to read and understand.

Its layout lacks visual hierarchy and consistency, as all charts have the same size, spacing is uneven, and colors are random, forcing users to search manually for important information.

The interface is static and non-interactive, offering no tooltips, filtering, or real-time updates, making the website useless for traders. Overall, it fails to guide the user's attention or provide feedback, violating key HCI principles such as clarity, consistency, and user control.

Why was it designed that way?

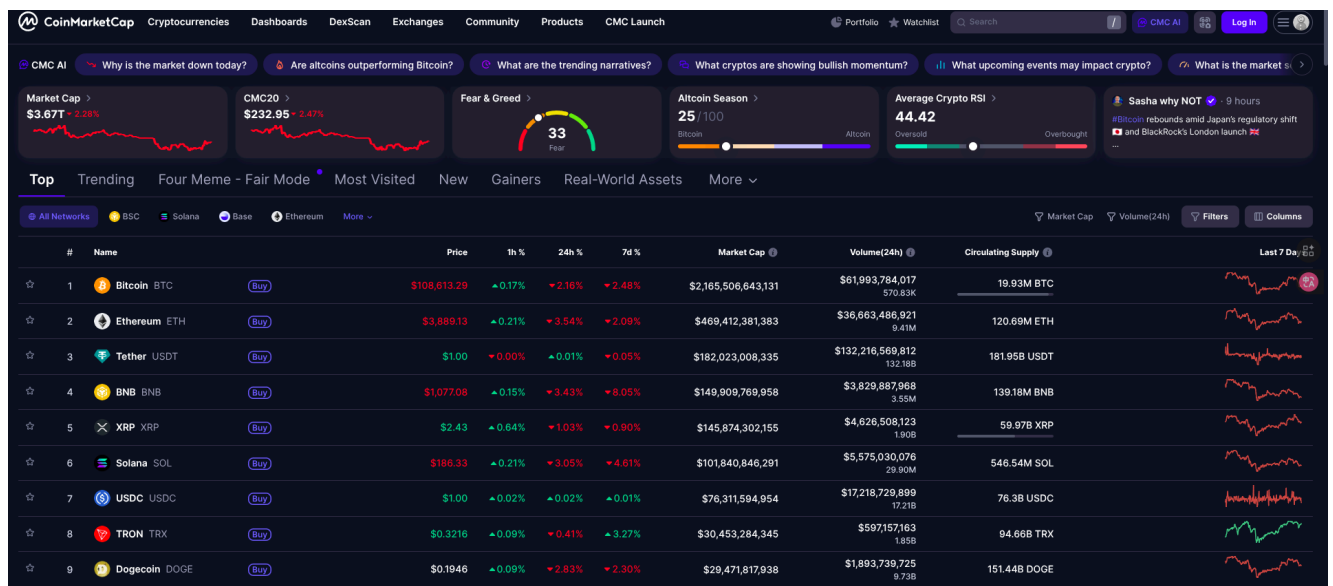
It is possible that the programmer didn't put too much effort into the assignment or had a tight deadline. The designer may have prioritized data retrieval and display rather than layout and interaction. The dashboard was probably designed to be displayed on TV, or other screens rather than on a desktop computer, that's why it's less interactive and static.

Possible improvements

- Round large numbers to two decimal places
- Add currency symbols and thousand separators for readability.
- Ensure consistent formatting across all data sources and charts.
- Resize and reorder components according to relevance
- Remove sidebar from chart
- Apply a coherent color palette
- Introduce real-time data updates
- Integrate mini-trend charts
- Fix the navbar bug
- Use consistent font sizes
- Add loading animations
- Include an interactive comparison tool
- Provide range time selection

Example of good design: CoinMarketCap

CoinMarketCap's main objective is to provide users with real-time information about cryptocurrencies, including prices, market capitalization, trading volume, and rankings. It aims to help investors, traders, and analysts easily compare and track digital assets across different exchanges and timeframes. Overall, the platform's goal is to make the global crypto market transparent, accessible, and understandable for both beginners and professionals.



Why is it good?

- Uses compact, modular cards arranged logically at the top.
- Data tables dominate the lower section - what users care about most
- prices and percentage changes are emphasized, smaller text supports them.
- Uses dark theme with clear color coding: green for gains, red for losses
- High information density but cleanly formatted: commas in numbers, proper currency symbols, aligned columns
- Visual balance between text and white space
- Percentage changes and trends are visually encoded
- Small chart inside table helps in tracking trend
- Rich interactivity - sorting, filtering, searching, hovering, and detailed coin info on click
- Dynamic updates and animations enhance engagement

CoinMarketCap is considered a good user interface because it successfully combines complex information with visual simplicity. It follows fundamental HCI principles such as consistency, feedback visibility, and aesthetic design. Users can perform their main tasks finding and comparing crypto assets quickly and effortlessly without being overloaded. Website looks professional and user friendly.